Projection of *Apis dorsata* Distribution in Southeast Asia Under Moderate (SSP245) and High (SSP585) Emissions by 2050

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I. INTRO

The giant honey bee (*Apis dorsata*) is a wild honey bee species, native to South and Southeast Asia. *Apis dorsata* plays an important ecological role as a pollinator. This species contributes to ecosystems by supporting plant reproduction and maintaining interactions between plants and other organisms. Through pollination networks, *Apis dorsata* helps maintain ecological stability and resilience of ecosystems.

Apis dorsata was chosen as the focal species because it is native to Southeast Asia, and gives a broader view of climate change study since this species is typically found in the wild. Unlike honey bee species like *Apis mellifera* that are usually maintained through human intervention, *A. dorsata* depends entirely on natural environments for nesting, foraging, and reproduction. Therefore, changes in its distribution pattern can provide important insights into how climate change may affect wild pollinator populations.

Species Distribution Modeling (SDM) is a useful approach to predict potential habitat suitability. This is done by relating species occurrence records with environmental variables. This model is meant to predict the distribution of *Apis dorsata* by the year 2050, by mapping suitability habitats under two different emission scenarios, SSP245 (moderate emissions) and SSP585 (high emissions). Shared Socioeconomic Pathway (SSP) is a set of future climate scenarios, commonly used in climate modeling to project future climate conditions. SSP245 is considered moderate, with the society continues developing but emissions do not decline quickly enough to avoid warming. SSP585 is considered as high emissions, with society relying heavily on fossil fuels and greater greenhouse gas emissions.

This independent study aims to identify potential areas of habitat loss and gain under future climate conditions, and provide insights into how climate change may influence the distribution of a key native pollinator species in Southeast Asia.

II. METHODS

2.1 SPECIES DISTRIBUTION MODELING (SDM)

A Species Distribution Modeling (SDM) was developed to estimate the current and future habitat suitability of *Apis dorsata.* Environmental variables were obtained from the WorldClim database using the 10 arc-minute spatial resolution dataset. Species occurrence records of *Apis dorsata* were obtained from the Global Biodiversity Information Facility (GBIF).

Occurrence records were cleaned prior to modeling to reduce spatial bias and improve model reliability. The cleaning process included species filtering (only records identified as *Apis dorsata*), coordinate validation and cleaning, and duplicate occurrence points removal to prevent overrepresentation of sampling locations.

Bioclimatic variables were prepared as predictor layers for the SDM. Future climate projections were obtained from the BCC-CSM2-MR climate model under the SSP245 (moderate) and SSP585 (high) emission scenarios. These projections were used to evaluate potential changes in *Apis dorsata* habitat suitability under future climate conditions.

2.2 BACKGROUND POINT GENERATION

Background points were generated from *Apis dorsata* occurrence data to represent available environmental conditions across the study area.

2.3 RANDOM FOREST MODELING

A Random Forest modeling was used to model the relationship between *Apis dorsata* occurrence records and environmental variables. The model was trained using occurrence points and generated background points. Model performance was evaluated based on its ability to distinguish suitable and unsuitable habitats.

2.4 MODEL EVALUATION

The Out-of-Bag (OOB) error rate was used to estimate model prediction error of the Random Forest model. Lower OOB error values indicate better model performance, suggesting that the model has a stronger ability to classify suitable and unsuitable habitats.

2.5 VARIABLE IMPORTANCE ANALYSIS

The contribution of each environmental variable to model predictions was assessed using variable importance measures. Mean Decrease Gini scores were used to determine the importance of predictor variables, where higher values indicated greater influence.

2.6 HABITAT SUITABILITY MAPPING

Suitability values were converted into binary habitat suitability groups using a threshold value of 0.5, where

Values $\geq$ 0.5 = suitable habitat

Values < 0.5 = unsuitable habitat

Habitat suitability was compared across three different conditions:

1. Current climate conditions
2. Future climate projection under SSP245
3. Future climate projection under SSP585

2.7 HABITAT CHANGE ANALYSIS

Changes in suitable habitat were evaluated by comparing future projections with current habitat suitability. Habitat change was classified as:

-1 = habitat lost (suitable habitat under current conditions becomes unsuitable in the future)

0 = no change (habitat suitability remains the same)

+1 = habitat gained (new suitable habitat appears under future conditions)

The percentage change in suitable habitat area was calculated using:

Habitat change (%) = (Future area - Current area) / Current area * 100

This calculation estimates the increase or decrease in suitable habitat availability.

III. RESULTS

A Random Forest species distribution model was developed for *Apis dorsata* using occurrence records from GBIF and four bioclimatic variables from WorldClim. The model achieved an out-of-bag accuracy of 85.94%.

Table 3.1. Confusion matrix results

Actual	Predicted Correctly	Total Samples	Accuracy
Background (0)	2.225	2.660	83.65%
Presence (1)	2.952	3.364	87.75%
Overall	5.177	6.024	85.94%

OOB estimate of error rate = 14.06%, accuracy = 85.94%

The confusion matrix showed that the model correctly classified 2.225 out of 2.660 background points and 2.952 out of 3.364 presence points. Overall, the model achieved an accuracy of 85.94%, meaning that approximately 86% of all observations were correctly classified. The OOB error rate was 14.06%, indicating that only a small proportion of samples were incorrectly predicted. The relatively low OOB error rate indicates that the model had a good predictive reliability, and could be used to project habitat suitability under current and future climate scenarios.

Table 3.2. Bioclimatic variables that affect habitat suitability for *Apis dorsata*

Variable	MeanDecreaseGini Value
BIO4 (Temperature Seasonality)	863.6
BIO15 (Precipitation Seasonality)	793.6
BIO1 (Annual Mean Temperature)	615.2
BIO12 (Annual Precipitation)	599.7

The results shown in the table indicated that temperature seasonality (BIO4) was the most influential predictor of *Apis dorsata* distribution in Southeast Asia, followed by precipitation seasonality (BIO15). Annual mean temperature (BIO1) and annual precipitation (BIO12) were also important, but contributed less to model performance. These results suggest that the species may be particularly sensitive to seasonal climatic variability rather than average climatic conditions alone. Future alterations in rainfall and temperature seasonality associated with climate change may therefore affect the distribution and migration patterns of this species.

Table 3.3. Number of suitable habitats (represented in cells) for *Apis dorsata* across Southeast Asia under 3 different climate (current, SSP245, and SSP585)

Scenario	**Cells**	**Change (%)**
Current	3.131	0
SSP245 Year 2050	1.621	-48.2
SSP585 Year 2050	1.801	-42.5

Table 3.4. Habitat change analysis by 2050 under moderate emissions (SSP245) and high emissions (SSP585)

Change Category	**Cells**	
	SSP245	**SSP585**
Habitat Lost	1.926	1.871
No Change	13.062	12.992
Habitat Gained	1.151	1.276
Net Change	-775	-595
Mean Suitability	0.264	0.273

Under the SSP245 scenario, suitable habitat decreased from 3.131 to 1.621 cells. This represents a 48.2% reduction in suitable habitat area. The habitat change analysis identified 1.926 cells as habitat loss, 1.151 cells as newly suitable habitat gain, and 13.062 cells showing no change. Overall, the model predicted a net habitat loss of 775 cells. The mean habitat suitability value under SSP245 was 0.264, indicating generally low suitability across projected landscapes.

Under the SSP585 scenario, suitable habitat decreased to 1.801 cells. This corresponds to a 42.5% reduction in suitable habitat area. The habitat change analysis identified 1.871 cells as habitat loss, 1.276 cells as newly suitable habitat gain, and 12.992 cells showing no change. The overall predicted net loss was 595 cells, with a mean suitability value of 0.273.

Both scenarios projected substantial habitat contraction for *Apis dorsata* by the year 2050. However, SSP585 retained a slightly greater area of suitable habitat than SSP245.

IV. DISCUSSION

The map (Figure I) suggests that current areas with high suitability for *Apis dorsata* are Thailand, Cambodia, Peninsular Malaysia, Sumatra, Java, Sulawesi, and Philippines. These are warm tropical areas with relatively stable climates and strong seasonal rainfall patterns. In comparison, areas with low suitability are Borneo and Papua. This is plausible because *Apis dorsata* is a migratory species and responds strongly to flowering cycles, which are influenced by seasonal climate patterns. Giant honey bees (*Apis dorsata)* typically prefer warm tropical climates and areas with seasonal flowering resources.

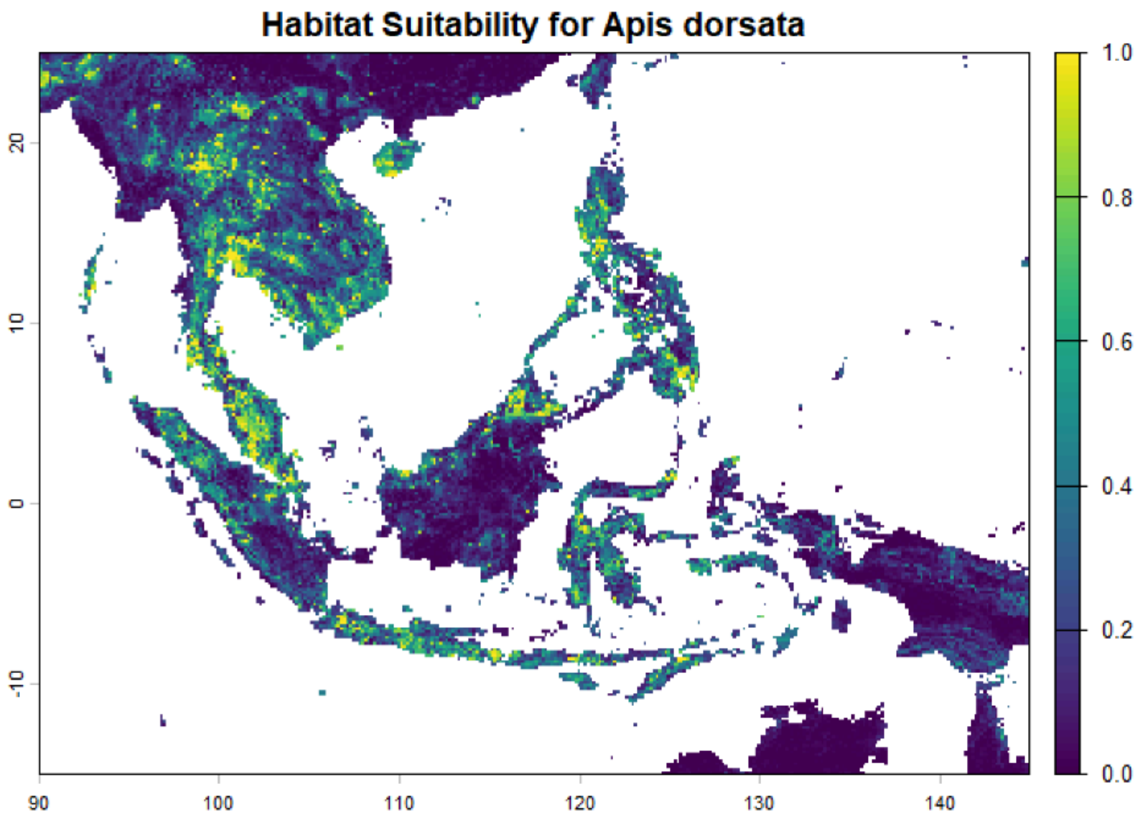

Figure 1. Current map of suitable habitats for *Apis dorsata* across Southeast Asia (high value = high suitability, low value = low suitability)

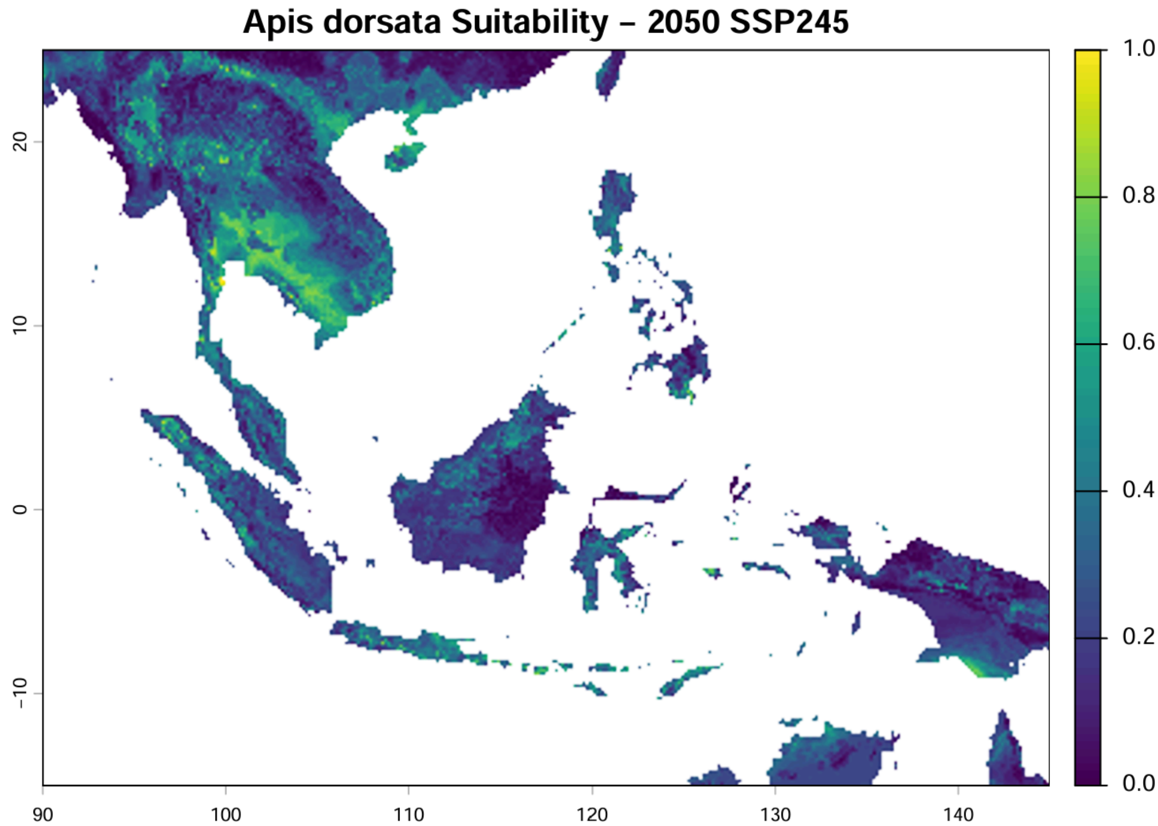

Figure 2. Projected habitat suitability map for *Apis dorsata* by 2050 under SSP245 (moderate emission) scenario

Projected areas with high suitability for *Apis dorsata* habitat by 2050 under SSP245 scenario were mainly distributed across tropical regions of Southeast Asia, including Papua, Java, Aceh, coastal area of Sumatra, Serawak, Thailand, and Cambodia (Figure 2). These regions represent areas where future climatic conditions remain favorable for *Apis dorsata*, likely due to persistence of suitable temperature and precipitation conditions. These results suggest that some regions may continue to support *A. dorsata* populations by 2050, while other areas may experience habitat contraction due to changing climate. Areas like Brunei, Peninsular Malaysia, and Philippines (Figure 1) showed a significant decrease in suitability by the year 2050.

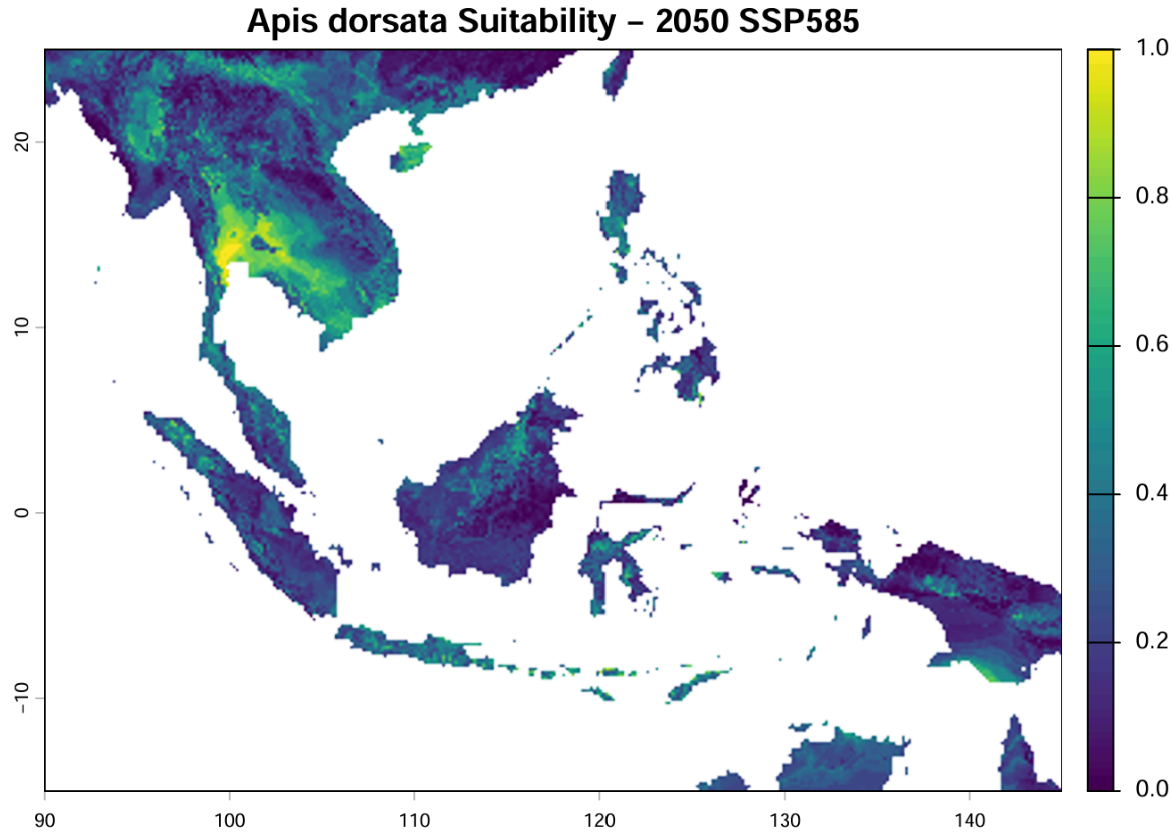

Figure 3. Projected habitat suitability map for *Apis dorsata* by 2050 under SSP585 (high emission) scenario

In contrast to the SSP245 scenario, Cambodia and Thailand showed a significant increase in habitat suitability under the SSP585 scenario (Figure 3). This suggests that despite the higher greenhouse gas emission, future climatic conditions in these regions may become more favorable for *Apis dorsata* compared with current conditions or the moderate emission scenario.

Compared to current climatic conditions, the SSP245 scenario for 2050 resulted in a projected reduction of approximately 48% in suitable habitats for *Apis dorsata* across Southeast Asia. While the SSP585 scenario resulted in 42% reduction. Although a stronger decline in habitat suitability might be expected under the SSP585 scenario due to its higher greenhouse gas emission, the results showed that SSP585 retained a slightly larger area of suitable habitat compared with SSP245.

That is plausible because out of all four most important ecological variables, seasonality plays a bigger role than mean climatic variables, indicating that changes in seasonal patterns may strongly influence future habitat suitability. It is possible that under the SSP585 scenario, some regions become warmer, but precipitation seasonality or temperature seasonality shift into more favorable conditions. These areas may act as potential refuges for the wild honey bees in the future. These findings highlight that stronger warming does not always translate into a decrease in habitat suitability, as species responses depend on the interaction between different environmental factors and regional climate patterns.

V. LIMITATIONS

Several limitations while conducting the study included:

5.1 Only climate variables were included

5.2 Land-use change, habitat fragmentation, and human disturbance were not considered

5.3 GBIF occurrence data may contain sampling bias

VI. REFERENCES

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